EEE202 ENGINEERS AND SOCIETY
ASSIGNMENT 2: INDUSTRIAL AND ENGINEERING SAFETY

1. Introduction

As a group of engineers, prepare a poster to help the society to understand the issues and mistakes in industrial and engineering safety, and preventive measures that can be undertaken for the future.

Transferable Skills: You must learn how to present to audiences whether you are presenting results from scientific research or conferences or presenting prototypes or final designs to potential clients. Posters are one type of presentation modes. They are an engaging, tangible way to walk others through your work by combining text, photos, diagrams, and graphs with **your accompanying verbal presentation**. In this activity, you are challenged to figure out how to communicate their work to an audience that could include experts, non-experts, and/or the general public.

2. Questions

Your poster must address and highlight the following statements.

Statement 1: "*Outright technical failure is rarely the sole cause of any disaster. Instead, disasters usually result from the cumulative effect of several shortcomings*."

a) Choose appropriate "industrial" accidents to illustrate Statement 1 and carefully analyse them under the following three headings:
   i) Design faults that contributed to the event
   ii) Human or operational errors that contributed to the event
   iii) Management and organisational problems that contributed to the event

Statement 2: "*The analysis of several accidents involving technology shows that history has a habit of repeating itself and that engineers are not learning from their mistakes*."

a) What conclusions can you draw? What lessons have been learned from these incidents for the future?
b) Do you agree or disagree with the statement, and if so, why?

3. Specific Instructions

a) Group of a maximum of four (4) members.
b) The role(s) of each member must be reported as a separate appendix.
c) The work must involve a literature review which requires e-library, e-databases, articles, etc. This poster's references must be included as a separate appendix.
d) Only the team leader is required to upload the poster, video and appendix to Moodle by 26th December 2022, Tuesday, 5:00 PM.
e) Late poster submission may result in a 20% penalty.
f) No submission will be accepted after a day of the deadline.
g) Any indication of unethical behaviour will be awarded "0".

4. Deliverables

a) A poster must be prepared to answers the questions in Section 2. It must be cited (in APA format) where appropriate.
b) A 2-minute video must be prepared to illustrate the poster.
c) An appendix outlining (1) each member's role(s) and (2) the references used in the poster.

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No.	Name	ID
1		
2		
3		
4		

	Evaluation Criteria	Exceptional (4)	Acceptable (3)	Acceptable (2)	Marginal (1)	Unacceptable (0)	Points
POSTER CONTENTS (70%)	Transfer	Makes explicit references to past accidents and applies in an innovative (new & creative) way that knowledge to demonstrate comprehension in the cumulative effect of a number of shortcomings.	Makes references to past accidents and shows evidence of applying that knowledge to demonstrate comprehension in the cumulative effect of a number of shortcomings.	Makes references to past accidents and attempts to apply that knowledge to demonstrate comprehension in the cumulative effect of a number of shortcomings.	Makes vague references to past accidents but does not apply knowledge to demonstrate comprehension in the cumulative effect of a number of shortcomings.		
	Reflection	Reviews past accidents in depth to reveal significantly changed perspectives about industrial safety, which provide foundation for expanded knowledge, growth, and maturity over time (i.e., safety awareness, prevention of accidents, and others).	Reviews past accidents in depth, revealing fully clarified meanings or indicating broader perspectives about industry safety.	Reviews past accidents with some depth, revealing slightly clarified meanings or indicating a somewhat broader perspectives about industry safety.	Reviews past accidents at a surface level, without revealing clarified meaning or indicating a broader perspective about industry safety.		
	Need for Life-Long Learning	Fully aware of external sources of material. Effective use of supplementary resources to justify points for discussion. Includes properly cited sources.	Reasonable awareness and use of external resources to explain points for discussion. Includes properly cited sources.	Some evidence of efforts to locate and use external resources to explain points for discussion. Includes few citations and few facts.	Little or no awareness and/or use of external sources of information. No citations are included.		

	Evaluation Criteria	Exceptional (4)	Acceptable (3)	Acceptable (2)	Marginal (1)	Unacceptable (0)	Points
*PRESENTATION (30%)	Organisation (Flow of Information) and Presentation	Introduction engages the interest of reader(s). Concisely presents the main points of the assignment and the characteristics of technical communication in a well-organized manner.	Introduction successfully frames the issues. Adequately presents the main points of the assignment in fairly well-organized manner.	Introduction outlines or lists arguments but does not generate interest. Presents the main points of the assignment, but they are not as well-organized.	Introduction minimally outlines arguments. Does not sufficiently present the main points of the assignment and is disorganized.	Introduction do little more than state the arguments.	
	Figures, Graphs and Tables (Effectiveness)	All figures and/or tables are effectively interpreted and discussed in the poster.	Most figures and/or tables are properly interpreted and important features noted.		Many figures and/or tables are not interpreted. Important features are not communicated or understood.	Figures and/or tables are not used effectively. Little understanding of important features or issues.	
	Visual Presentation	Overall visually appealing; not cluttered; colours and patterns enhance readability; uses font sizes which facilitate the organization, presentation, and readability of the work; graphics are engaging and enhance the text content; and contents are clearly arranged so that the viewer can understand order without narration.	Overall visually appealing; not cluttered; colours and patterns support readability; adequate use of font sizes to facilitate the organization, presentation, and readability of the work; graphics enhance the text content; and contents are arranged as such that the viewer can understand order without narration.	Visual appeal is adequate; somewhat cluttered; colours and patterns detract from readability; use of font sizes to facilitate the organization, presentation, and readability of the work is somewhat inconsistent/distracting; graphics adequately enhance the text; and contents arrangement is somewhat confusing.	Not very visually appealing; cluttered; colours and patterns hinder readability; use of font sizes to facilitate the organization, presentation, and readability of the research is inconsistent/distracting; graphics do not enhance the text; and the content arrangement is somewhat confusing.		
						Penalty (if rules violated)	
						Total Points	

*Weightage is subjected to changes

Comments: